

JAMES GRAY

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EDUCATION

University of Warwick – BEng Computer Systems Engineering (Year in Industry)

Sept. 2022 – June 2026

- **Grade: First (81.1%)**
- **Key Concepts:** AI, ML, Neural Computing, Operating Systems, Networks, Computer Architecture, FPGAs & SoCs, Compilers, Databases, Signal Processing, Analogue Electronic Design, Maths

Reading School

Sept. 2020 – July 2022

- **Grades: A*A*A*** (Maths, Physics, Computer Science)

SKILLS & AWARDS

- **Languages & Technologies:** C, C++, C#, Python, Java, Haskell, MATLAB, TypeScript, JavaScript, SQL, HTML, CSS, React, PyTorch, Verilog, HLS, Docker, Linux
- **Awards:** Exceptional Achievement in Second Year, Computer Science Department Award

WORK EXPERIENCE

Qube Research & Technologies – Quantitative Technology Intern

June 2024 – Present

- Designed and built a **C++ process-management service** that lets support teams control and monitor system processes via a UI and code, eliminating manual SSH interaction.
- Developed a tool that **auto-generates environment-specific configurations** and wires them into the process-management service, establishing a single source of truth.
- Reworked the **CI/CD pipeline** to support **C# builds and tests on Linux**, raising code coverage and unblocking Linux developers.
- Built an **LLM support assistant** (React.js) on a **ReAct loop** with custom tools for multi-step reasoning, context access, and output verification, improving answer accuracy.

MyTutor – Tutor

Jan. 2023 – Present

- Communicated complex concepts simply to help **A-Level and GCSE students** master subject material.

KEY PROJECTS

Deep Learning for Efficient Sleep Staging with PPG Data

(Python, PyTorch, Deep Learning, Time-Series)

- Pinpointed the **architectural bottlenecks** driving inference cost in state-of-the-art PPG AI sleep-staging models.
- Designed and implemented **Mamba** and **efficient-attention** architectures in PyTorch, which held accuracy ($\kappa = 0.74$) across **3 clinical datasets**.
- Raised inference throughput **3.8x** while cutting **VRAM 53%** and **model size 57%**, making deployment far more scalable.

Conjugate Gradient Optimisation on a 3D Mesh

(C, AVX-256, OpenMP, Cache Optimisation)

- Optimised a 3D-mesh conjugate gradient solver with **AVX-256**, **OpenMP**, and locality refactoring for an **8.88x speedup**.

Mini C Compiler

(C++, LLVM, Compilers)

- Built a **top-down recursive-descent** lexer and parser in C++, generating code via **LLVM IR**, for a subset of C.
- Produced **colourised, informative error diagnostics** more detailed than mainstream C compilers.

Multithreaded Packet Sniffer

(C, Networking, Multithreading)

- Developed a sniffer to detect domain blacklist violations, SYN floods, and ARP cache poisoning.
- Implemented a thread-safe packet queue with **pthreads** feeding a thread pool, processing **1,000,000s of packets without loss**.
- Validated for memory leaks and race conditions with **Valgrind**, **Helgrind**, and **GDB**.

FPGA Pacman

(Verilog, Vivado, Nexys 4 FPGA)

- Implemented Pacman's game logic (movement, collisions, and scoring) as **hardware state machines**, taking the **top score in the year**.
- Rendered the game from sprites in on-chip **BRAM**, with custom frame-drawing logic and a real-time **VGA** driver built in hardware.

Other Projects

- Resistor Image Scanner (Python, OpenCV, K-means, Flask), Magnetic Electron Trap simulation (Python, SciPy, 20x speedup), Movie Information Viewer (Java, from-scratch data structures), Gig Management System (Java, PostgreSQL), Discord Drive cloud storage (Python, Quart), Rhythm-game Discord bot, Maze & Circuit Solvers, Line-Following Buggy (C), Self-Balancing Robot (Simulink).